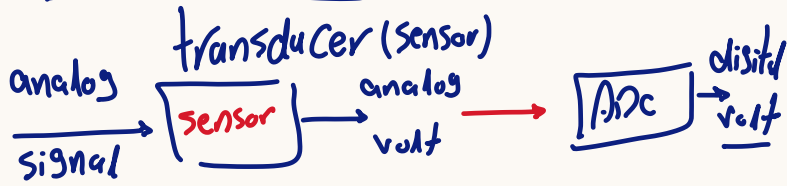


transducers

→ MC! Really analog volt → analog volt → [ADC] → digital volt



temp sensor

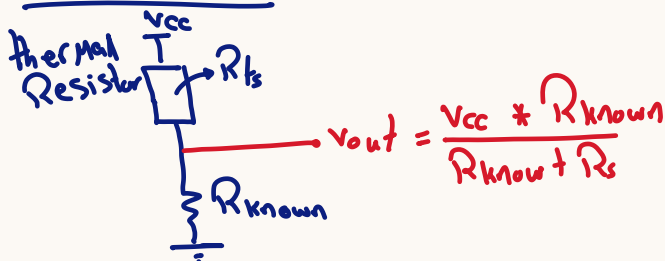
* How sensor convert from analog signal to analog volt?
 signal → Resistance → volt

→ ex: temp-sensing

① Resistor → the value of Resistor Change by temp



② volt divider



temp → Resistance → voltage

thermistors

NTC
 negative temp
 Coefficient

$$R \propto \frac{1}{T}$$

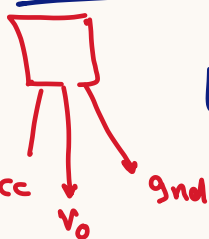
(-60 : 170)

PTC
 Positive temp
 Coefficient

$$R \propto T$$

< -60 > 150

* sensor LM35 | NTC | single ended



LM35 $\propto$ Vout
 0°C → 0V
 1°C → 10mV

→ ① digital volt
 → ② analog volt
 → ③ temp

$$LM35 \propto \frac{1}{R_t}$$

$$V_{out} \propto \frac{1}{R_t}$$

$$uint16_t \text{ analog_volt} = (\text{digital} \times \text{step})$$

$$= (\text{digital} \times V_{ref} \times 1000) / 1024;$$

$$\text{analog_volt} = (uint16_t) ((uint32_t) \text{digital} \times V_{ref} \times 1000) / 1024;$$

$$temp = \frac{\text{analog_volt} (m)}{10}$$

1 → 10mV
 ? → 200

LDR! light dependent Resistor

